

**GREENOPS**  
**IPS12 – INTELLIGENT DATA CENTER RESOURCE**  
**OPTIMIZATION AND SUSTAINABILITY MANAGEMENT**  
**PLATFORM**

**PHASE I PROJECT REPORT**

*Submitted by*

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*in partial fulfillment for the award of the degree*

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### **BONAFIDE CERTIFICATE**

Certified that this project report titled “**IPS12 – INTELLIGENT DATA CENTER RESOURCE OPTIMIZATION AND SUSTAINABILITY MANAGEMENT PLATFORM**” is the bonafide work of AISHWARYA A (230701015), AISHWARYA M (230701016), KAVIYA MADHIRAJU (230701148), and KARTHIKHA SRE M (230701143), who carried out the project work under my supervision. Certified further that, to the best of my knowledge the work herein does not form part of any thesis or dissertation on the basis of which degree or award was conferred on an earlier occasion on this or any other candidates.

**This project addresses the following Sustainable Development Goal(s):**

**Goal No : 13 - Climate Action - Take urgent action to combat climate change and its impacts by reducing emissions and strengthening climate resilience**

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## ABSTRACT

Data centers consume vast electricity and water, with significant waste from idle servers, stale or duplicate storage, and unoptimized cooling. Existing sustainability tools often monitor resources in isolation and report metrics without simulating the impact of optimization actions before execution. GreenOps bridges this gap by ingesting multi-domain telemetry (compute, power, cooling, storage) to compute live metrics like PUE, WUE, energy, cost, and carbon. A statistically derived rule engine identifies idle servers and storage inefficiencies using server-type-specific thresholds. Four machine learning models support the platform: two validation models trained on 10,000 telemetry records estimate CPU utilization and power draw, while two forecasting models predict near-term workload demand. Before proposing server consolidation, GreenOps executes a what-if simulation evaluating post-move CPU, memory, network, and thermal headroom against live and forecasted limits. Recommendations are ranked by weighted sustainability and risk factors, logged for auditing, and restricted via role-based access control. By keeping final execution decisions under human operator control, GreenOps delivers transparent decision support for data center sustainability. This report details the architecture, modelling, recommendation logic, and experimental validation of the platform.

***Keywords: GreenOps, Data Center Sustainability, Resource Optimization, Telemetry Analytics, Machine Learning, Workload Forecasting, What-If Simulation, PUE, WUE, Server Consolidation***

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## LIST OF ABBREVIATIONS

|                |                                      |
|----------------|--------------------------------------|
| PUE            | Power Usage Effectiveness            |
| WUE            | Water Usage Effectiveness            |
| CPU            | Central Processing Unit              |
| kWh            | Kilowatt-hour                        |
| kg             | Kilogram                             |
| MAE            | Mean Absolute Error                  |
| RMSE           | Root Mean Squared Error              |
| R <sup>2</sup> | Coefficient of Determination         |
| ML             | Machine Learning                     |
| API            | Application Programming Interface    |
| REST           | Representational State Transfer      |
| JWT            | JSON Web Token                       |
| RBAC           | Role-Based Access Control            |
| ESG            | Environmental, Social and Governance |
| NIC            | Network Interface Card               |
| SLA            | Service Level Agreement              |
| PDU            | Power Distribution Unit              |
| ORM            | Object-Relational Mapping            |
| CRUD           | Create, Read, Update, Delete         |

# CHAPTER 1

## INTRODUCTION

### 1.1 GENERAL

Data centers now account for a rapidly growing share of global electricity demand and industry studies consistently find that a significant fraction of that consumption is not doing useful work: servers sit idle at low utilization while still drawing near-full power, storage volumes accumulate stale and duplicated data that is never accessed again and cooling systems run without any real-time awareness of the load they are actually cooling. This waste is not exotic or hard to find; it is observable in ordinary infrastructure telemetry but it is rarely acted upon because the information needed to see it is scattered across separate monitoring systems for compute, power, cooling and storage and because most existing tools stop at reporting a dashboard number rather than proposing and safety-checking a concrete action.

GreenOps is a full-stack observability and recommendation platform built to address this gap directly. It structures every telemetry stream through the same five-stage pipeline: observing, analyzing, predicting, evaluating and recommending. The system never executes an infrastructure change on its own. It detects waste using transparent rule-based logic, validates current server behavior using machine learning models trained on real historical telemetry and forecasts near-term demand using models trained on the system's own accumulated history, simulates the safety and impact of a proposed consolidation before it happens and hands a ranked, itemized recommendation to a human operator who makes the final call. Every decision the operator takes is logged with the exact estimate that was shown at the time, producing an auditable trail suitable for sustainability and ESG reporting.

## 1.2 PROBLEM STATEMENT

Operators of data center infrastructure need a way to keep answering a few practical questions on an ongoing basis. Which servers are actually idle and worth consolidating? Which stored data is stale or duplicated and could be archived or deduplicated? Which storage volumes are over-provisioned compared to what they actually hold? And if a workload is moved onto a candidate server, is that move actually safe given the server's current and near future load, memory, network and thermal headroom? Is the data center sustainable? How much water is being used and how much carbon is being emitted? What to do to make it more sustainable?

Most existing tools don't answer all of this. They usually cover only one piece, compute or power or cooling or storage in isolation. They apply a single flat threshold no matter what type of hardware is involved and they don't simulate what would actually happen before recommending an action. Most tools don't tell us what's a directly measured value, what's a modelled estimate, and what simply isn't supported yet. That distinction matters a lot once the output is being used to justify a real infrastructure change or reported externally as a sustainability number.

## 1.3 OBJECTIVES

- To ingest and validate telemetry across four domains (server compute, power, cooling and storage) on a continuous, per-server basis.
- To compute live, formula-based sustainability metrics (PUE, WUE, energy, cost and carbon) and clearly separate genuinely measured ratios from labelled estimates.
- To detect operational waste, including idle servers, stale data, duplicated data and over-provisioned storage, using a rule engine with



per-server-type thresholds derived statistically from real historical data instead of one flat threshold applied to every server type.

- To train, compare and honestly report the accuracy of machine learning models that estimate CPU utilization and power draw from environmental and workload features, used as an explainable validation layer.
- To forecast near-term server workload at multiple horizons, using whichever method fits each horizon rather than forcing a single model onto every time range.
- To simulate, before any recommendation is shown, whether consolidating an idle server onto a candidate target would be safe both right now and in the near future, across CPU, memory, network and thermal dimensions.
- To rank pending recommendations using a configurable, weighted combination of estimated energy, cost, carbon and water impact against operational risk.
- To keep the human operator as the sole decision-maker: every decision and its supporting estimate is permanently logged for audit and ESG reporting and role-based access control separates who may decide from who may attest execution.

## **1.4 SCOPE OF THE PROJECT**

The current implementation covers the complete pipeline from telemetry ingestion through to ranked, human approved recommendations and an ESG decision log. The system runs against a simulated fleet of thirteen servers spanning four server types (Compute, GPU, Storage and Edge), three cooling types (Air, Hybrid and Liquid) and three regions (APAC, EU and Middle East), with several servers in each type deliberately biased toward lower utilization so the recommendation engine always has more than one genuine candidate to work with.

Telemetry for all four domains (compute, power, cooling and storage) is produced by purpose built simulators rather than read from physical hardware, since production data center access was not available for this project. These simulators were built to mirror real monitoring behavior closely, including realistic posting intervals, range validated readings and duplicate delivery handling, so that the ingestion and analysis layers behave exactly as they would against a live facility. The two estimation models (CPU and power) are trained offline on a real, static 10,000 row “Green AI Data Center Telemetry” dataset. The two forecasting models (CPU and memory) are trained on telemetry accumulated by the live simulator itself.

The project covers rule based waste detection across all four categories (idle servers, stale data, duplicated data and over provisioned storage), with per type thresholds derived statistically. It covers a four dimension consolidation safety check spanning CPU, memory, network and thermal headroom, evaluated both against a server's current state and its near term forecast. It covers a priority weighted recommendation engine whose scoring weights an operator can adjust live and a role based access system that separates who may approve a recommendation from who may later confirm it was carried out. It also covers a dedicated ESG reporting layer that aggregates approved decisions into a facility level sustainability record, downloadable as CSV or PDF.

Cost and carbon figures throughout the system are computed using placeholder tariff and grid intensity constants, whether it is a predicted power draw, an expected saving or a water usage figure, is clearly distinguished in the interface from a value that was actually measured. That distinction was treated as a core design requirement throughout the project, since a system meant to eventually support real sustainability reporting has to be honest about which of its numbers can be trusted at face value and which are a considered estimate.

## **CHAPTER 2**

### **LITERATURE SURVEY**

Research on data center efficiency has developed along a few separate paths that rarely meet. One path optimizes where workloads run. Another applies machine learning to predict resource use before or after the fact. A third looks at carbon-aware scheduling and cooling. A fourth is less concerned with optimization and asks instead whether sustainability is even being measured and reported at all. The study of these existing approaches provides valuable insights into the development of the proposed GreenOps system.

#### **2.1 LITERATURE REVIEW**

Ahmed, Brown and Gegov investigated dynamic resource allocation through workload prediction in “Dynamic Resource Allocation through Workload Prediction for Energy-Efficient Computing” [1]. Their work examined how workload prediction can support dynamic resource allocation and improve energy efficiency while maintaining service requirements. This approach demonstrates the value of predictive resource management. However, it does not combine workload prediction with a unified set of sustainability metrics, recommendation generation, what-if simulation and human approval.

Alex, Ojo and Awuor proposed a carbon-aware and energy-efficient virtual machine placement approach using deep Q-networks and agglomerative clustering [2]. Their work considers carbon emissions, energy efficiency and service-level agreement requirements when determining virtual machine placement. Although this approach supports sustainability-oriented resource allocation, its primary focus is placement optimization rather than an integrated workflow for telemetry monitoring, waste detection, forecasting, recommendations and operator decision-making.

Bawa, Kaur Chahal and Kaur proposed an ensemble learning approach for cloud resource management and workload prediction using XGBoost and LightGBM [3]. Their study investigated workload activity classification and the prediction of CPU and memory utilization. The findings highlight the usefulness of ensemble learning for anticipating resource requirements. However, prediction remains the primary outcome and is not directly connected to a sustainability-aware recommendation mechanism for selecting corrective actions.

Beloglazov, Abawajy and Buyya, in “Energy-Aware Resource Allocation Heuristics for Efficient Management of Data Centers for Cloud Computing,” proposed heuristics for virtual machine placement and migration to improve data center energy efficiency [4]. Their work demonstrated the value of dynamic consolidation for reducing power consumption compared with static resource allocation. However, the approach focuses on automated resource management and does not provide the same operator-reviewed recommendation workflow proposed in GreenOps.

Deva and Jain investigated cloud resource utilization prediction using Azure Automated Machine Learning and XGBoost in enterprise environments [5]. Their work explores automated model selection and machine-learning-based prediction of cloud resource utilization. This demonstrates how machine learning can support resource planning and utilization analysis. However, resource prediction alone does not provide a complete workflow for assessing sustainability impacts, recommending corrective actions and obtaining operator approval.

The European Commission has established corporate sustainability reporting requirements that require applicable organizations to disclose environmental and sustainability-related information [6]. These requirements emphasize the

importance of systematic sustainability measurement and transparent reporting. However, reporting frameworks primarily address disclosure obligations rather than providing continuous infrastructure monitoring, machine-learning-based prediction or operational recommendations.

Gorey discussed the land and water implications associated with the rapid expansion of artificial intelligence infrastructure [7]. The discussion highlights how the growth of data centers can place additional pressure on natural resources and the regions hosting these facilities. This broader environmental perspective shows why data center sustainability should not be assessed through electricity consumption alone. However, the work does not propose an operational mechanism for continuous resource monitoring, prediction or optimization.

Grochulska-Salak et al. examined sustainable data centers through the use of renewable energy and green-blue infrastructure [9]. Their work considers several sustainability dimensions, including energy efficiency, water usage, waste heat recovery, renewable energy integration and land use. This demonstrates the importance of assessing data center sustainability through multiple indicators. However, a broad sustainability framework does not itself implement a continuous machine-learning-based system for detecting inefficiencies and recommending operational changes.

Khan, Tian, Ilager and Buyya investigated workload forecasting and energy-state estimation in cloud data centers using multiple resource characteristics, including provisioned resources, disk throughput and network activity [10]. Their work highlights the importance of considering different resource parameters when estimating workload behaviour and energy-related characteristics. However, forecasting and estimation do not by themselves

provide an integrated workflow for waste detection, sustainability assessment, recommendation generation and operator approval.

Lovász, Niedermeier and de Meer studied the performance implications of energy-aware virtual machine consolidation in “Performance Tradeoffs of Energy-Aware Virtual Machine Consolidation” [11]. Their work considered the trade-off between energy savings and system performance, showing that consolidation should not be evaluated only in terms of energy reduction. This is important because aggressive consolidation can affect performance. However, the study primarily focuses on consolidation trade-offs rather than a continuous system that combines forecasting, sustainability assessment and operator-reviewed recommendations.

Radovanović et al. presented “Carbon-Aware Computing for Datacenters,” which describes an approach for shifting flexible workloads across time and geographical locations according to forecasted grid carbon intensity [13]. This work demonstrates how carbon emissions can be considered alongside computing requirements when scheduling workloads. However, this approach depends on workload flexibility and access to infrastructure across different locations, which may not be available in smaller or single-site data center environments.

Subramanian, Vasan and Sivasubramaniam investigated server consolidation in “Reducing Data Center Power with Server Consolidation: Approximation and Evaluation” [17]. They treated server consolidation as an optimization problem similar to bin packing, aiming to reduce power consumption while satisfying resource requirements. However, the approach does not integrate machine-learning-based workload forecasting with broader sustainability metrics and operator-oriented decision support.

Tao and Gao examined the environmental and public-health implications associated with global data center expansion and highlighted the need for further empirical investigation into these impacts [18]. Their work broadens the discussion of data center sustainability beyond operational energy consumption to include potential environmental and societal consequences. However, the study focuses on broader environmental impacts rather than providing a practical system for operational resource optimization and machine-learning-based decision support.

Tripathi et al., in research conducted by the Council on Energy, Environment and Water and SYSTEMIQ, examined the development of India's data center ecosystem and its associated infrastructure, energy, resource and resilience requirements [19]. Their work highlights the importance of planning for the growing demands of data center infrastructure. However, ecosystem-level analysis does not directly provide an operational system for detecting resource inefficiencies and recommending corrective actions within individual data centers.

Uddin et al. investigated server consolidation as a technique for improving cloud data center power efficiency and reducing overall cost of ownership [20]. Their work examines the relationship between consolidation, power consumption and operational costs. However, the primary focus is on evaluating consolidation strategies rather than continuously detecting inefficiencies, forecasting future demand and guiding operators through a structured decision-making process.

Vaccaro, Tagliabue and Aldinucci investigated sustainable data centers with an emphasis on energy efficiency and resource optimization [21]. Their work contributes to the broader study of improving resource utilization and sustainability in data center environments. However, the study's broad

sustainability and optimization perspective does not itself provide the complete integrated monitoring-to-decision workflow proposed in GreenOps.

Zhan et al. investigated data center cooling system optimization using offline reinforcement learning [23]. Their approach uses historical operational data to learn cooling-control policies without relying on direct trial-and-error experimentation on live infrastructure during training. The study demonstrates the potential of reinforcement learning for cooling optimization, including its application in a production data center. However, its primary focus is cooling control rather than combining workload forecasting, resource waste detection, multiple sustainability metrics and human-reviewed recommendations in one system.

From the reviewed literature, several research directions can be identified. Existing studies have addressed workload prediction and intelligent resource allocation [1]–[5], sustainability measurement and environmental impact [6], [7], [9], [18], [19], workload forecasting and energy-state estimation [10], energy-aware consolidation [11], [17], [20], carbon-aware scheduling [13], and sustainable resource optimization [21], [23]. However, these capabilities are generally addressed separately.

The major research gap identified from the literature is the limited integration of continuous telemetry monitoring, rule-based waste detection, machine-learning-based prediction, sustainability assessment, recommendation generation, what-if simulation and human-reviewed decision-making within a unified operational workflow. Existing approaches often focus on a particular optimization task, prediction model or sustainability assessment. GreenOps addresses this integration gap by connecting these capabilities while retaining the final operational decision with the human operator.



Thus, the reviewed literature establishes the importance of intelligent resource management, predictive modelling and comprehensive sustainability measurement in data centers. GreenOps builds on these research directions by combining waste detection, prediction, sustainability assessment, recommendation generation and what-if simulation into a unified decision-support platform.

## 2.2 SUMMARY OF LITERATURE SURVEY

**Table 2.1 Comparative Analysis of Existing Literature**

| Ref. | Study        | ML / Optimization Technique Used            | How the Model / Technique Was Used                                                                       | Main Focus                                        | Limitation / Gap Relative to GreenOps                                                                                               |
|------|--------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| [1]  | Ahmed et al. | Neural-network-based workload prediction    | Predicts workload behaviour to support dynamic resource allocation and improve energy efficiency.        | Energy-efficient dynamic resource allocation      | Does not combine prediction with unified sustainability metrics, recommendations, what-if analysis and human approval.              |
| [2]  | Alex et al.  | Deep Q-Network and Agglomerative Clustering | Uses clustering and deep reinforcement learning to guide carbon-aware and energy-efficient VM placement. | Carbon-aware VM placement                         | Focuses on VM-placement optimization rather than an integrated monitoring, forecasting, recommendation and human-decision workflow. |
| [3]  | Bawa et al.  | XGBoost and LightGBM ensemble learning      | Uses ensemble models to identify workload activity and predict CPU and memory utilization.               | Workload prediction and cloud resource management | Prediction is the main outcome. It does not connect predictions to sustainability-aware recommendations or what-if analysis.        |

|     |                         |                                            |                                                                                                              |                                                     |                                                                                                                                                           |
|-----|-------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| [4] | Beloglazov et al.       | Adaptive VM-consolidation heuristics       | Detects overloaded and underloaded hosts, migrates VMs, consolidates workloads and powers down idle servers. | Energy-aware VM allocation and server consolidation | Does not provide the integrated forecasting, sustainability assessment, what-if analysis and human-reviewed recommendation workflow proposed in GreenOps. |
| [5] | Deva and Jain           | Azure AutoML and XGBoost                   | Uses Azure AutoML for model selection and XGBoost for predicting cloud resource utilization.                 | Cloud resource-utilization prediction               | Focuses on prediction and does not extend the results into integrated sustainability assessment, recommendations and human-reviewed decisions.            |
| [6] | European Commission     | Sustainability reporting framework         | Establishes requirements for reporting environmental and sustainability-related information.                 | Corporate sustainability reporting                  | Focuses on disclosure and reporting rather than real-time monitoring, prediction or operational recommendations.                                          |
| [7] | Gorey                   | Environmental and resource-impact analysis | Examines land and water implications associated with the expansion of AI infrastructure.                     | Land and water impact of AI infrastructure          | Highlights resource impacts but does not provide an operational mechanism for monitoring, prediction or optimization.                                     |
| [9] | Grochulski-Salak et al. | Sustainability indicators and framework    | Considers energy efficiency, renewable energy, water use, waste heat, carbon emissions and land use.         | Holistic data center sustainability                 | Provides a broad sustainability perspective but does not implement a continuous ML-based operational optimization system.                                 |

|      |                    |                                                                             |                                                                                                                                   |                                                   |                                                                                                                                                     |
|------|--------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| [10] | Khan et al.        | Linear Regression, Ridge, ARD, ElasticNet, GRU and clustering               | Uses forecasting models for workload prediction and clustering for VM energy-state estimation.                                    | Workload forecasting and energy-state estimation  | Does not integrate forecasting with waste detection, unified sustainability metrics, recommendations, what-if simulation and operator approval.     |
| [11] | Lovász et al.      | Energy- and performance-aware consolidation algorithms                      | Evaluates VM consolidation while considering both energy savings and possible performance degradation.                            | Energy versus performance trade-off               | Focuses on consolidation trade-offs rather than continuous prediction, sustainability assessment or recommendation generation.                      |
| [13] | Radovanić et al.   | Carbon-intensity forecasting, demand prediction and risk-aware optimization | Uses forecasts and optimization to shift flexible workloads toward lower-carbon periods and locations.                            | Carbon-aware workload scheduling                  | Depends on workload flexibility and distributed infrastructure. Does not focus on local resource waste detection or human-reviewed recommendations. |
| [17] | Subramanian et al. | Approximation algorithm                                                     | Models server consolidation as a variable-sized bin-packing problem to reduce the number of active servers and power consumption. | Power reduction through server consolidation      | Does not use ML forecasting or integrate broader sustainability metrics with operator decision support.                                             |
| [18] | Tao and Gao        | Environmental impact analysis                                               | Examines environmental and public-health implications associated with data center expansion.                                      | Environmental and societal impact of data centers | Focuses on broader environmental impacts rather than operational resource optimization or ML-based decision support.                                |

|      |                 |                                                     |                                                                                                                                 |                                                           |                                                                                                                                                                                    |
|------|-----------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [19] | Tripathi et al. | Industry and sustainability analysis                | Examines infrastructure, energy, resource and resilience requirements associated with India's growing data center ecosystem.    | Data center infrastructure , energy and resilience        | Provides ecosystem-level analysis rather than an operational system for detecting and resolving resource inefficiencies.                                                           |
| [20] | Uddin et al.    | Server-consolidation techniques                     | Evaluates consolidation strategies and their effects on power efficiency and total cost of ownership.                           | Energy consumption, power efficiency and operational cost | Primarily evaluates consolidation outcomes rather than providing a continuous detection, prediction, recommendation and decision workflow.                                         |
| [21] | Vaccaro et al.  | Sustainability and resource-optimization approaches | Examines approaches for improving energy efficiency and resource utilization in sustainable data centers.                       | Sustainable data center operation                         | Does not implement the complete ML-based monitoring-to-decision workflow proposed in GreenOps.                                                                                     |
| [23] | Zhan et al.     | Physics-informed offline reinforcement learning     | Learns cooling-control policies from historical operational data and applies the learned policy to optimize cooling operations. | Data center cooling optimization                          | Focuses specifically on cooling control rather than integrating workload forecasting, resource waste detection, broader sustainability metrics and human-reviewed recommendations. |

From the comparative analysis, existing studies primarily focus on individual aspects of sustainable data-center management such as resource allocation, server consolidation, workload prediction, carbon-aware scheduling, cooling optimization or sustainability measurement. Although several studies employ

machine learning or intelligent optimization techniques, these capabilities are generally addressed independently. Limited attention is given to integrating continuous telemetry monitoring, waste detection, workload forecasting, sustainability assessment, recommendation generation and what-if simulation into a single operator-oriented workflow. GreenOps addresses this gap by connecting these capabilities while retaining human oversight over the final operational decision.

## **CHAPTER 3**

### **SYSTEM ANALYSIS AND DESIGN**

#### **3.1 OVERVIEW OF THE PROPOSED SYSTEM**

GreenOps watches four streams of data center telemetry, uses simple explainable rules to spot waste, uses two families of machine learning models to forecast and validate server behaviour, simulates the consequence of consolidating workloads before anything happens, and hands a human operator a ranked list of recommendations with honest, itemized impact estimates. The system never executes anything itself.

#### **3.2 SYSTEM ARCHITECTURE**

The architecture is organized into four layers: a data layer of independent telemetry simulators, a FastAPI backend that performs ingestion, storage, metrics calculation, rule-based waste detection, ML validation, forecasting, what-if simulation, and recommendation ranking; a PostgreSQL persistence layer; and a React frontend that exposes the results through role-appropriate dashboards. Figure 3.1 shows the complete architecture, from the four simulated monitors on the left, through the backend's seven internal stages, to the frontend pages on the right and the database tables at the bottom.

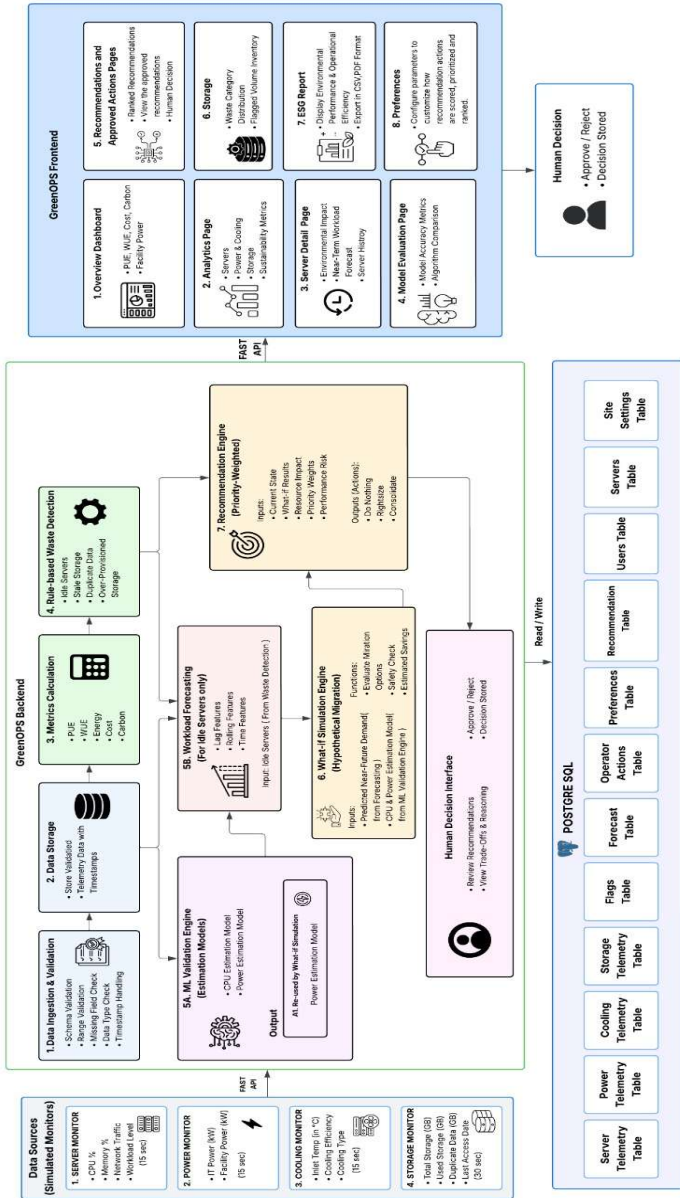


Fig 3.1 GreenOps System Architecture

At the center of the backend, the Rule-based Waste Detection module and the Metrics Calculation module both read from validated, stored telemetry and feed two parallel downstream paths: an ML Validation Engine (offline-trained CPU and power estimation models) and a Workload Forecasting module restricted to servers already flagged idle. Both paths feed a What-if Simulation Engine, which in turn feeds the priority-weighted Recommendation Engine. Every recommendation the engine produces is surfaced to a Human Decision Interface, and the operator's approve/reject/snooze decision is written back to the database, closing the loop.

### **3.3 SYSTEM DESIGN**

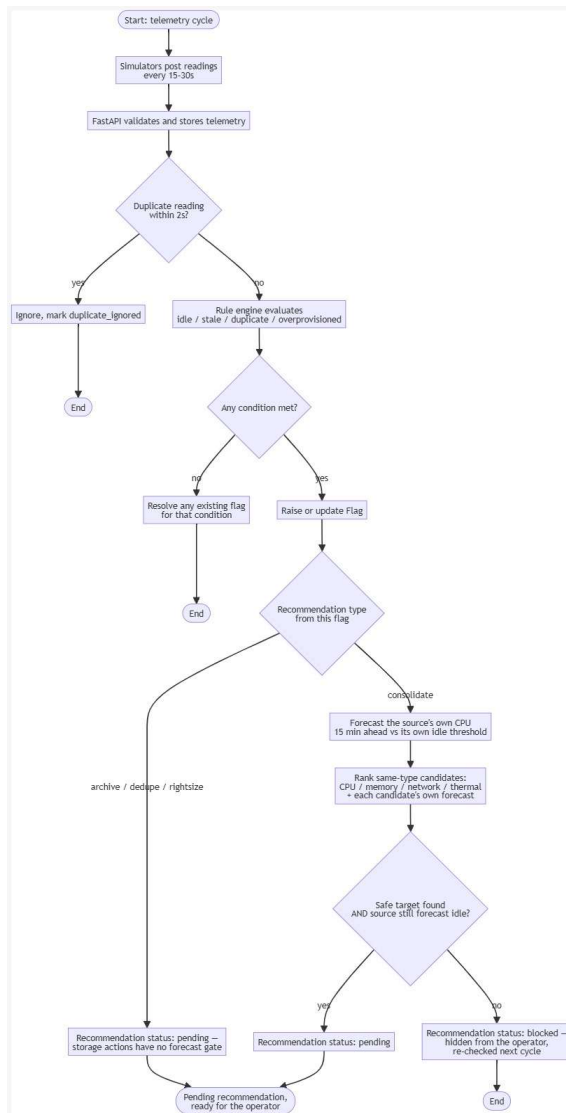
#### ***3.3.1 ACTIVITY DIAGRAM***

The activity diagram shown in Figure 3.2 illustrates the continuous monitoring and analysis workflow executed by the GreenOps telemetry pipeline. The process initiates with the telemetry cycle, where independent simulator scripts post structured readings every 15 to 30 seconds. Upon ingestion, the FastAPI backend performs schema validation and range checks. To ensure data integrity, any duplicate payload received within a 2-second sliding window is automatically dropped and marked as ignored.

Validated telemetry flows into the rule-based waste detection engine, which evaluates the latest observations against domain-specific thresholds for idle servers, stale data, duplicate data, and over-provisioned storage. If no waste condition is met, any previously active flag for that condition is resolved. When a condition is triggered, a new flag is raised or updated. For non-compute waste flags (archive, deduplicate, rightsize), recommendations are directly marked as pending. For idle server flags, the pipeline initiates a 15-minute workload forecast for the source server. If a safe candidate target exists and the source server is projected to remain idle, a consolidation recommendation is queued for



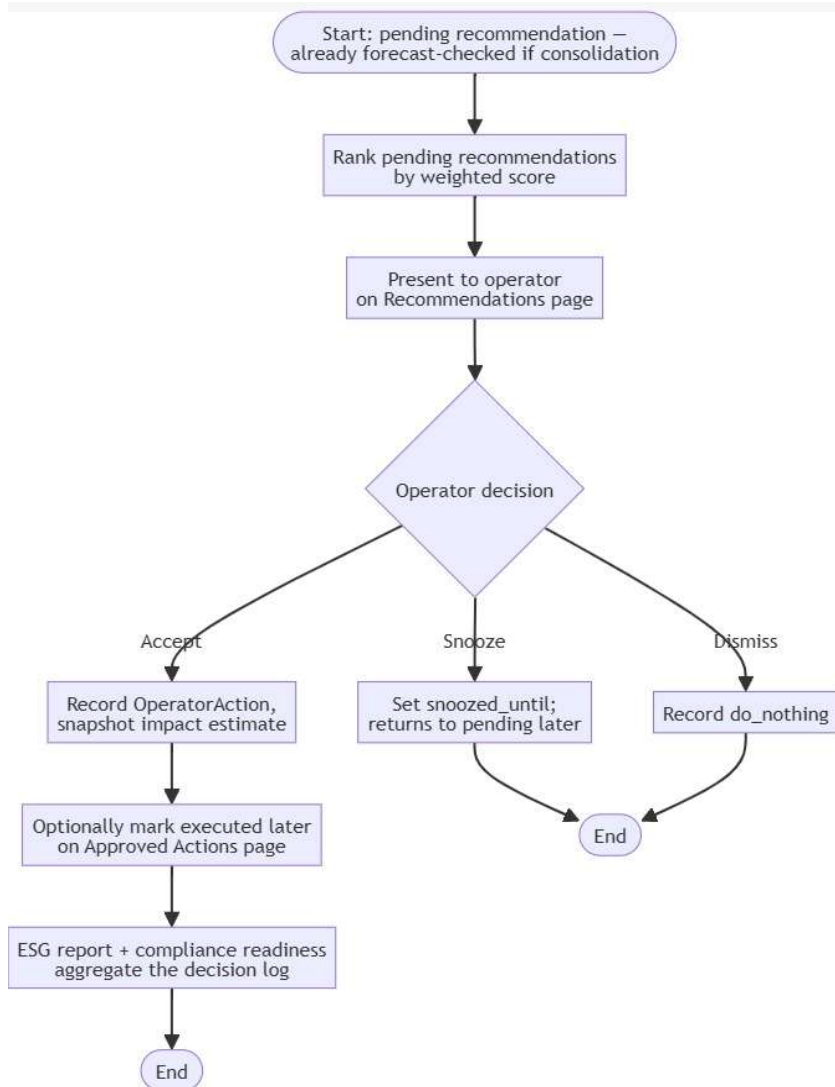
operator review; otherwise, the recommendation is temporarily blocked and re-evaluated during the next cycle.



**Fig 3.2 Activity Diagram for Monitoring and Analysis**

Figure 3.3 details the decision-making and execution workflow for pending recommendations. Pending actions - already validated for safety in the case of server consolidations - are first prioritized and ranked by the recommendation engine using the system's weighted scoring matrix across energy, carbon, cost, water, and operational risk metrics. The ranked list is then presented to the operator through the Recommendations dashboard interface.

The human operator evaluates the presented options and selects one of three primary decision paths: Accept, Snooze, or Dismiss. Accepting a recommendation records an OperatorAction audit log entry, freezing a snapshot of the estimated savings at decision time. The accepted action can later be manually attested as executed on the Approved Actions page, allowing the ESG and compliance modules to aggregate the verified decision history. Selecting Snooze assigns a deferred timestamp, after which the recommendation automatically reverts to pending status. Dismissing a recommendation records a do-nothing action, concluding the lifecycle for that specific evaluation cycle.



**Fig 3.3 Activity Diagram for Recommendation and Action**

### 3.3.2 ER DIAGRAM

Figure 3.4 shows the Entity-Relationship (ER) diagram for the GreenOps platform, which maps out the core database tables and how they connect. The **SERVER** entity sits at the center, holding each server's identifier, type, cooling type and datacenter region, and every other operational table points back to it through `server_id`. Each server posts readings to four telemetry tables: **SERVER\_TELEMETRY** for CPU, memory and network metrics, **POWER\_TELEMETRY** for IT and facility power, **COOLING\_TELEMETRY** for inlet temperature and cooling efficiency, and **STORAGE\_TELEMETRY** for total, used and duplicate data volumes. These links are one-to-many and optional, so a server can be registered before any data arrives and then build up a time-stamped history. The **FORECAST** table also ties to **SERVER**, storing predicted CPU values for a chosen horizon next to the measured ones.

The rule engine reads this telemetry and raises **FLAG** entries against the affected server, each with a type, severity, metric value and resolution state. Every flag produces a **RECOMMENDATION** with a priority, a workflow status and an optional snooze date, ranked using the energy, cost, carbon, water and risk weights in **PREFERENCES**. **SITE\_SETTINGS** supplies the wider context, such as the grid emission factor, installed IT capacity and renewable energy factor, which feed the estimates attached to each decision. Operators, stored in the **USER** entity, review each recommendation and choose what to do. Each decision is saved in **OPERATOR\_ACTION** with the action taken, the servers involved, estimated savings in energy, carbon and cost, and the deciding user, giving a full audit trail from raw data to the final human decision.



### 3.4 DATA LAYER - TELEMETRY SIMULATION

Nothing in the running system reads real hardware. Four independent Python scripts each generate one telemetry domain and POST it to the FastAPI backend every fifteen seconds, per server, continuously, standing in for real monitoring integrations that were not available for this project.

**Table 3.1 Telemetry Simulators and Their Output Fields**

| Simulator          | Posts to           | Fields                                                                           |
|--------------------|--------------------|----------------------------------------------------------------------------------|
| server_monitor.py  | /telemetry/server  | cpu_utilization, memory_utilization, network_throughput_gbps, workload_intensity |
| power_monitor.py   | /telemetry/power   | it_power_kw, facility_power_kw                                                   |
| cooling_monitor.py | /telemetry/cooling | inlet_temperature_c, cooling_efficiency, cooling_type                            |
| storage_monitor.py | /telemetry/storage | total_storage_gb, used_storage_gb, duplicate_data_gb, last_accessed_days_ago     |

Thirteen simulated servers (S1–S13) exist, each with a fixed identity assigned at registration and never changed at runtime, spanning four server types, three cooling types, and three regions, with a deliberately varied workload bias so that some servers are realistically idle and others realistically busy.

**Table 3.2 Simulated Server Identity and Workload Bias**

| Server | Type    | Cooling | Region | Workload bias    |
|--------|---------|---------|--------|------------------|
| S1     | Compute | Air     | APAC   | 0.65             |
| S2     | Compute | Air     | APAC   | 0.60             |
| S3     | GPU     | Hybrid  | EU     | 0.30             |
| S4     | Storage | Liquid  | ME     | 0.15 (idle)      |
| S5     | Edge    | Air     | APAC   | 0.12 (idle)      |
| S6     | Compute | Air     | APAC   | 0.05 (very idle) |
| S7     | GPU     | Hybrid  | EU     | 0.55             |
| S8     | GPU     | Hybrid  | ME     | 0.10 (idle)      |

| Server | Type    | Cooling | Region | Workload bias |
|--------|---------|---------|--------|---------------|
| S9     | Storage | Liquid  | ME     | 0.45          |
| S10    | Storage | Liquid  | APAC   | 0.14 (idle)   |
| S11    | Edge    | Air     | APAC   | 0.50          |
| S12    | Edge    | Air     | EU     | 0.16 (idle)   |
| S13    | Compute | Air     | ME     | 0.70          |

Each simulator generates its readings from a small, explicit formula rather than pure random noise, so that later modules can be verified by hand. `server_monitor.py` draws a workload value around each server's fixed bias, and derives CPU, memory, and network from it, so memory and network carry no information CPU does not already contain. `power_monitor.py` reads each server's live CPU and memory over the API before generating its own reading, so that IT power is a real function of load, and the ratio of facility power to IT power (PUE) is a real function of the server's live cooling efficiency-both of these were previously pure random draws and were fixed during this project. `cooling_monitor.py` similarly reads each server's live CPU and applies a per-cooling-type temperature-rise and efficiency-drop curve, so that air-cooled servers heat up fastest and lose efficiency fastest under load, hybrid sits between, and liquid barely moves; this causal direction runs only from load to environment, never the reverse, which is a known and explicitly documented limitation. `storage_monitor.py` generates capacity, staleness, and duplication metadata.

FastAPI validates every incoming reading against a Pydantic schema, writes it to the database with a timestamp, and silently drops an exact-duplicate delivery from the same server arriving within two seconds, so that a network retry never double-counts a reading.

### 3.5 RULE-BASED WASTE DETECTION ENGINE

Four checks, each simple enough to verify by hand, run continuously against the latest stored telemetry.

**Table 3.3 Rule Engine Flags**

| Flag            | Rule                                                  |
|-----------------|-------------------------------------------------------|
| idle_server     | 6-hour average CPU below that server type's threshold |
| stale_data      | last_accessed_days_ago > 90                           |
| duplicate_data  | duplicate_data_gb ÷ total_storage_gb > 20%            |
| overprovisioned | used_storage_gb ÷ total_storage_gb < 30%              |

The idle threshold is not a single flat number applied to every server. Per-type thresholds are derived once from the real **“Green AI Data Center Telemetry”** dataset as

$$\text{threshold} = \text{mean\_cpu}(\text{type}) - 1.5 * \text{std\_cpu}(\text{type})$$

clamped to the range [5%, 35%], with a severity weight scaled to that type's average power draw relative to the lowest-power type. This means an idle GPU server, which draws roughly 66% more power on average than an idle Edge server at the same CPU percentage in the real dataset, is flagged as materially higher-priority waste rather than identically to any other idle server.

**Table 3.4 Per-Type Idle Thresholds and Severity Weights**

| Server type | Idle threshold | Severity weight |
|-------------|----------------|-----------------|
| GPU         | 16.9%          | 1.66×           |
| Compute     | 17.3%          | 1.22×           |
| Storage     | 16.8%          | 1.06×           |
| Edge        | 18.9%          | 1.0× (baseline) |



A flag that stops being true is resolved on the same cadence it was raised, so nothing accumulates indefinitely; the synchronization routine also withdraws any pending recommendation whose underlying flag is no longer active, so a server flapping between idle and healthy does not pile up stale duplicate recommendations.

### 3.6 SUSTAINABILITY METRICS

Table 3.5 lists every sustainability metric the system computes, together with an honest statement of how trustworthy each one is—a distinction the system is deliberately explicit about throughout, rather than presenting every number with equal apparent confidence.

**Table 3.5 Sustainability Metrics and Trust Level**

| Metric          | Formula                                                                                         | Trust level                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| PUE             | $\text{facility\_power\_kw} \div \text{it\_power\_kw}$                                          | Genuine ratio of two directly simulated values                                          |
| Energy (kWh)    | $\Sigma (\text{Power} * \text{Elapsed Time})$                                                   | Corrected; previously over-counted $\approx 4\times$ by assuming one reading per minute |
| Cost            | $\text{energy\_kwh} \times \text{Tariff}$                                                       | Estimate                                                                                |
| Carbon          | $\text{energy\_kwh} \times \text{Carbon Factor}$                                                | Estimate                                                                                |
| Total water (L) | Cumulative $\text{energy\_kwh} \times \text{WUE factor}$ by cooling type, since telemetry began | Estimate, running total                                                                 |
| WUE (L/kWh)     | $\text{Water Consumption} \div \text{Energy}$                                                   | Estimate, genuine ratio in shape                                                        |

Water and carbon figures use published industry-average placeholders (WUE factors of 0.3, 1.8, and 0.9 L/kWh for Air, Hybrid, and Liquid cooling respectively) keyed only by the categorical cooling type, never by the live numeric cooling-efficiency reading. In a production deployment, cost, carbon, and WUE would be replaced by a real electricity tariff, a live grid carbon-intensity feed, and a real facility water meter.

### 3.7 MACHINE LEARNING VALIDATION MODELS

Two regression models are trained once, offline, on a real, static, 10,000-row Kaggle dataset (“Green AI Data Center Telemetry”)-never on live simulator output, so the two pipelines share only a schema, never an origin. Shared preprocessing scales numeric features and one-hot-encodes categorical features with the first category dropped, which is necessary because, without dropping it, one-hot-encoding every category alongside an intercept term creates a perfectly collinear feature matrix, a bug that was found and fixed during development.

#### 3.7.1 CPU ESTIMATION MODEL

**Target:** `cpu_utilization`.

**Inputs:** `workload_intensity`, `memory_utilization`, `network_throughput_gbps`, `inlet_temperature_c`, `cooling_efficiency`, `pue`, plus the categorical features `server_type`, `cooling_type`, `datacenter_region`, and `time_of_day`.

The model deliberately excludes its own target from its inputs to avoid leakage.

#### 3.7.2 POWER ESTIMATION MODEL

**Target:** `power`, rescaled from the dataset's 28–233 kW range down to the live simulated 1.2–4.5 kW range-a pure unit change, confirmed by identical  $R^2$  before and after rescaling.

Inputs are the same six numeric and four categorical features as the CPU model, plus `cpu_utilization`, which is legitimate here since CPU is not the target for this model.

#### 3.7.3 MODEL SELECTION RULE

Three candidate algorithms-Linear Regression, Random Forest (200 estimators), and XGBoost (200 estimators, depth 5) are trained for each target on an 80/20 split. The lowest mean absolute error wins by default, but a more complex

model (Random Forest or XGBoost) must beat Linear Regression by at least 5% relative MAE improvement to be selected over it, a promotion bar that was fixed before results were seen. On this dataset, Linear Regression won both targets detailed results are reported in Chapter 5.

## **3.8 WORKLOAD FORECASTING**

The forecasting module answers one question per tier: given a server's history, what will its CPU likely be? Two structurally different forecasting jobs exist and are not interchangeable.

### ***3.8.1 OPERATOR-FACING FORECAST***

Gated to servers currently flagged idle, this forecast covers three horizons, each handled by the method appropriate to its range rather than one method forced onto all three. The one-hour horizon uses its own trained lagged-feature model (five-, ten-, fifteen-, and twenty-minute CPU lags, the rolling one-hour average, memory, workload intensity, network throughput, and time-of-day encoded as sine/cosine), falling back to short-horizon trend extrapolation until at least thirty real lagged samples exist. The six-hour and twenty-four-hour horizons deliberately use no trained model at all: there is not yet enough real history at that range, and because each server's workload is generated from a fixed bias that never drifts or cycles, the honest estimate that far out is reversion to the server's own recent average rather than a straight-line extrapolation, which would badly overshoot. The long-horizon forecast explicitly refuses to answer - returning **“not enough history”** rather than a guess-unless real telemetry coverage reaches at least half the requested window with at least ten samples.

### ***3.8.2 CONSOLIDATION SAFETY FORECAST***

A completely separate, fifteen-minute-horizon model, used internally only by the what-if simulation described in Section 3.8, and always run on any

registered server with no idle gate-since a consolidation target is, by definition, not idle, and could never pass an idle-only gate. This forecast reuses the same CPU-lag architecture as the operator-facing model, plus a second, independently trained memory forecast model, since the safety check needs both dimensions.

Live prediction errors for any of these models will typically run higher than the reported offline metrics, because the models were trained on real historical relationships while live data comes from simulators using simpler generating formulas-a genuine, expected train/inference gap rather than a defect.

### 3.9 RECOMMENDATION ENGINE AND WHAT-IF SIMULATION

Each active flag is mapped to a recommended action type.

**Table 3.6 Flag-to-Recommendation Mapping**

| Flag            | Recommended action |
|-----------------|--------------------|
| idle_server     | consolidate        |
| stale_data      | archive            |
| duplicate_data  | deduplicate        |
| overprovisioned | rightsize          |

For idle\_server flags specifically, an actual what-if simulation is run rather than a source-only guess. Every same-type server that is not itself idle-flagged is considered a candidate. For each candidate, post-move CPU, memory, and network are computed by adding the source server's own load, discounted by a 0.9 overhead factor, onto the candidate's current load. Each candidate is then checked across four safety dimensions, all of which must pass for the candidate to be considered safe right now.

**Table 3.7 Consolidation Safety Check Dimensions**

| Dimension | Limit                                 | Basis                                                       |
|-----------|---------------------------------------|-------------------------------------------------------------|
| CPU       | < 75%                                 | Safety limit constant                                       |
| Memory    | < 75%                                 | Safety limit constant                                       |
| Network   | < 18.75 Gbps                          | 75% of an assumed 25 Gbps NIC                               |
| Thermal   | Predicted post-move inlet temp < 30°C | Reuses the cooling simulator's own load→temperature formula |

Independently of the current-state check, the candidate's own near-term CPU and memory forecast (15 minutes ahead, Section 3.7.2) is checked against the same limits. A candidate only counts as fully safe when both the current-state check and the forecast check pass - this catches a target that looks fine right now but is about to get busy on its own. Network and thermal have no forecast counterpart, since no trained model exists for either, so they remain current-snapshot checks only. The top-ranked safe candidate (safe first, then most remaining headroom) becomes the recommended target; if no candidate qualifies, the recommendation is explicitly marked unsafe with a stated reason, never silently approved, and all considered candidates (up to five) are exposed in the response so an operator can see what else was evaluated. The impact estimate for a recommended consolidation prefers the trained power model's prediction for the target's full post-move profile, scaled to facility power by the target's current PUE, falling back to a simpler CPU-ratio formula only if the model or the target's cooling telemetry is unavailable; every use of the power model in an estimate is stated explicitly, including the caveat that it is a legitimate estimate rather than a measurement.

Pending recommendations across all flag types are ranked by:

$$\text{score} = \sum(\text{weight}_i \times \text{normalized\_impact}_i) - \text{risk\_weight} * \text{risk}$$

normalized against the maximum of each dimension within the current pending batch so that kWh, currency, kilograms, and litres do not dominate each other purely by raw magnitude. The weights are stored in a single-row, organization-configurable Preferences table editable from the frontend; changing them re-ranks recommendations instantly without retraining anything. Risk for a consolidation recommendation is

$$\text{max}(\text{post\_move\_cpu}, \text{post\_move\_memory}) \div 75$$

for storage-only actions, which have no target server and no power effect, risk is a fixed 0.1.

### 3.10 OPERATOR DECISIONS AND AUDIT TRAIL

An operator with the `infrastructure_manager` role can consolidate (or the generic `rightsize` verb, for archive/deduplicate actions), snooze, or dismiss any pending recommendation. Every decision snapshots the what-if impact exactly as it was shown at the moment of the decision into an `OperatorAction` record-the ESG report reads this frozen snapshot, not a live recalculation, so the log always reflects exactly what the operator actually saw when deciding. Snoozing sets a `snoozed_until` timestamp; once that passes, the recommendation quietly returns to pending rather than being lost.

The Approved Actions page is the one narrow exception to the rule that GreenOps has no way to verify anything actually happened: an operator can manually attest, after the fact, that a consolidation was carried out. This is attestation, not verification-there is still no automatic confirmation mechanism anywhere in the system-and the ESG report that aggregates these records is explicitly a decision log, not a verified-savings report.

### 3.11 ROLE-BASED ACCESS CONTROL

Three roles are enforced both server-side, via a dependency checked on every mutating endpoint, and client-side, via route guards and conditional navigation.

**Table 3.8 Role-Based Access Control Matrix**

| Role                   | Can do                                                                                     |
|------------------------|--------------------------------------------------------------------------------------------|
| infrastructure_manager | Everything-accept/snooze/dismiss recommendations, edit preferences, manage users           |
| sustainability_manager | Read-only-dashboards and the ESG report                                                    |
| operations_engineer    | Mark approved actions as executed; cannot make the original accept/snooze/dismiss decision |

Authentication is JWT-based, with users seeded through a dedicated script.

### 3.12 DATABASE DESIGN

PostgreSQL is used in deployment, with SQLite available as a zero-code-change default controlled by a single environment variable. Schema changes are managed through Alembic, baselined onto the pre-existing schema. Eleven tables make up the schema: **users, servers (static metadata), four raw telemetry tables (server\_telemetry, power\_telemetry, cooling\_telemetry, storage\_telemetry), forecasts (a persisted forecast audit log), flags, recommendations, operator\_actions, and preferences.**

## CHAPTER 4

### IMPLEMENTATION

#### 4.1 TECHNOLOGY STACK

- **Backend:** Python, FastAPI, Pydantic (validation), SQLAlchemy (ORM), Alembic (migrations).
- **Database:** PostgreSQL in deployment, SQLite for local development, selected via the DATABASE\_URL environment variable.
- **Machine learning:** scikit-learn (Linear Regression, Random Forest, preprocessing pipeline), XGBoost.
- **Frontend:** React, Vite, react-router-dom, recharts.
- **Authentication:** JSON Web Tokens (JWT).
- **Simulation:** independent Python scripts per telemetry domain, run as long-lived processes..

#### 4.2 SYSTEM REQUIREMENTS

##### 4.2.1 HARDWARE REQUIREMENTS

**Table 4.1 Hardware Requirements**

| Component | Minimum Recommendation    | Purpose                                                                |
|-----------|---------------------------|------------------------------------------------------------------------|
| Processor | Dual-core 2 GHz or better | Runs the backend, frontend, and simulators.                            |
| Memory    | 8 GB RAM                  | Supports PostgreSQL, Python packages, browser, and model training.     |
| Storage   | At least 5 GB free space  | Stores source code, dependencies, database, logs, and model artifacts. |



|         |                          |                                                                   |
|---------|--------------------------|-------------------------------------------------------------------|
| Network | Localhost network access | Allows frontend, backend, and simulator processes to communicate. |
| Display | 1280 × 720 or better     | Recommended for dashboard and analytics review.                   |

#### 4.2.2 SOFTWARE REQUIREMENTS

**Table 4.2 Software Requirements**

| Software               | Recommended role                          |
|------------------------|-------------------------------------------|
| Python 3.10 or later   | Backend, simulators, training, and tests. |
| Node.js and npm        | React/Vite frontend development.          |
| PostgreSQL 14 or later | Production-style relational database.     |
| Git                    | Source-code version control.              |
| Visual Studio Code     | Local development environment.            |
| Modern web browser     | Dashboard access and UI testing.          |

The project is installed locally using the following commands:

```
None
cd greenops
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r backend/requirements.txt
cd frontend
npm install
```

On Windows PowerShell, the Python environment is activated using:

```
None
.venv\Scripts\Activate.ps1
```

## 4.3 DATABASE, REACT FRONTEND, TELEMETRY, AND API INTEGRATION

### DATABASE CONFIGURATION

PostgreSQL is created using the following SQL commands:

SQL

```
CREATE DATABASE greenops; CREATE USER greenops_user WITH PASSWORD 'change_this_password';  
GRANT ALL PRIVILEGES ON DATABASE greenops TO greenops_user;
```

The backend reads the database connection from an environment variable.

SQL

```
DATABASE_URL="postgresql+psycopg2://greenops_user:change_this_password@localhost:5432/greenops"
```

The main database tables store users, server metadata, server telemetry, power telemetry, cooling telemetry, storage telemetry, forecasts, flags, recommendations, operator actions, and priority preferences. Raw telemetry and decision records are stored in the database, while serialized machine-learning model files are stored as artifacts rather than as binary database columns.

SQLAlchemy maps Python classes to database tables. A telemetry record is first validated by Pydantic, then converted into a SQLAlchemy object, added to the active session, committed, and returned as a JSON response. Alembic is used to manage controlled schema changes when the database structure evolves.

### REACT AND VITE FRONTEND

The React interface is built with Vite. Vite provides the development server, module bundling, fast refresh, and production build process. In the existing project, the frontend is already initialized; its dependencies are installed with:

None

```
cd frontend  
npm install
```

```
//The frontend development server is started using:  
npm run dev
```

The frontend displays values supplied by the backend rather than independently recalculating authoritative metrics. This keeps the backend as the source of truth and avoids inconsistent values between screens. Recharts is used to display time-series trends, server comparisons, utilization changes, energy patterns, and forecast results.

## TELEMETRY SIMULATOR EXECUTION

Four independent Python processes generate the telemetry used during the demonstration. They are started in separate terminals after the backend is available.

```
None  
python simulators/server_monitor.py  
python simulators/power_monitor.py  
python simulators/cooling_monitor.py  
python simulators/storage_monitor.py
```

## API INTEGRATION

The API is divided into functional groups. Telemetry endpoints receive records from the simulators. Server endpoints return current state and historical records. Analytics endpoints calculate time-window summaries. Forecast endpoints return future workload estimates. Recommendation endpoints return ranked actions and what-if details. Operator-action endpoints store human decisions. Report endpoints return ESG and decision-log information.

**Table 4.3 Primary REST Endpoint Reference**

| Endpoint                          | Purpose                                                               |
|-----------------------------------|-----------------------------------------------------------------------|
| POST /telemetry/server            | Ingest server telemetry                                               |
| POST /telemetry/power             | Ingest power telemetry                                                |
| POST /telemetry/cooling           | Ingest cooling telemetry                                              |
| POST /telemetry/storage           | Ingest storage telemetry                                              |
| POST /servers/register            | Register a new server before it can post telemetry                    |
| GET /servers                      | Current state of all servers                                          |
| GET /servers/{id}/detail          | Combined live view for one server, used by the Server Detail page     |
| GET /servers/{id}/history         | Historical telemetry for one server                                   |
| GET /servers/{id}/prediction      | ML CPU and power estimate vs measured                                 |
| GET /servers/{id}/forecast        | One-hour, six-hour or twenty-four-hour CPU projection                 |
| GET /forecasts                    | Persisted forecast history and validation metrics                     |
| GET /flags                        | All active rule-based flags                                           |
| GET /storage/flags                | Storage-specific flags only                                           |
| GET /thresholds                   | Per-server-type idle thresholds and severity weights                  |
| GET /analytics/summary            | PUE, WUE, energy, cost and carbon overview                            |
| GET /analytics/daily              | Daily time-series data behind the Analytics page's charts             |
| GET /analytics/history            | Longer-range historical analytics data                                |
| GET /recommendations              | Ranked pending recommendations                                        |
| GET /recommendations/{id}         | Full detail for one recommendation, without re-running the simulation |
| GET /recommendations/{id}/what-if | Consolidation safety and impact simulation                            |
| POST /recommendations/{id}/action | Record the operator's accept, snooze or dismiss decision              |
| GET /operator-actions             | Decision history for the Approved Actions page                        |
| PATCH /operator-actions/{id}      | Attest that an approved action was executed                           |
| GET /preferences                  | Read the current recommendation-ranking weights                       |

| Endpoint                                                       | Purpose                                                     |
|----------------------------------------------------------------|-------------------------------------------------------------|
| PUT /preferences                                               | Update the recommendation-ranking weights                   |
| GET /reports/esg                                               | Aggregated ESG summary over a date range                    |
| GET /reports/esg/export                                        | ESG report as CSV                                           |
| GET /reports/esg/export/pdf                                    | ESG report as PDF                                           |
| GET /model/evaluation                                          | Three-model comparison metrics for CPU and power estimation |
| POST /auth/login                                               | Authenticate and receive a session token                    |
| GET /auth/me                                                   | Return the currently authenticated user's role and details  |
| GET /users, POST /users, PATCH /users/{id}, DELETE /users/{id} | Manage user accounts and roles, infrastructure manager only |

#### 4.4 MACHINE-LEARNING, FORECASTING, RECOMMENDATION, AND DECISION IMPLEMENTATION

The machine-learning layer contains two related but separate tasks. The CPU estimation model estimates the current CPU utilization from current workload and environmental features. The workload-forecasting model predicts a future CPU or workload value using timestamped lag features. The CPU model is not silently renamed as the forecasting model; the two tasks have different targets and input structures.

The offline training process loads the historical telemetry dataset, handles missing values, encodes categorical features, scales numeric features where required, constructs the training matrix, and splits the data chronologically. Linear Regression, Random Forest, and XGBoost are compared using MAE, RMSE, and  $R^2$ . The selected model and its evaluation metrics are saved for later inference.

A forecast generated at 10:00 for 10:15 is not considered validated immediately. When telemetry for 10:15 becomes available, the predicted value is compared

with the actual value. The system can then calculate the absolute error, squared error, and aggregate MAE or RMSE over many forecasts.

The recommendation engine begins with active flags. An idle-server flag may produce a consolidation recommendation, stale data may produce an archive recommendation, duplicate data may produce a deduplication recommendation, and over-provisioned storage may produce a rightsizing recommendation. For consolidation, the what-if engine evaluates candidate target servers by adding the source workload with an overhead factor. It checks CPU, memory, network, and thermal safety limits in both the current state and the near-term forecast state.

The recommendation score combines normalized energy, cost, carbon, water, and risk impacts using configurable weights. The recommended action is not automatically executed. The human decision interface allows an authorized operator to approve, snooze, dismiss, or reject the recommendation. The system stores the selected decision, user identity, timestamp, score, candidate details, and impact snapshot in an audit record.

## **CHAPTER 5**

### **RESULTS AND DISCUSSION**

#### **5.1 RULE ENGINE RESULTS**

With the thirteen simulated servers running continuously, the rule engine correctly identifies the servers deliberately biased toward idle (S4, S5, S6, S8, S10, S12 in Table 3.1) within the first few minutes of telemetry collection and correctly assigns the higher severity weight to idle GPU servers (S8) relative to idle servers of other types at a comparable CPU percentage, matching the per-type thresholds derived in Section 3.4. Flags resolve automatically once a server's rolling average CPU rises back above its type's threshold and no duplicate recommendation accumulates when a server flaps between idle and healthy across consecutive evaluation windows, confirming the intended behaviour of the flag-synchronization routine.

#### **5.2 ML MODEL EVALUATION RESULTS**

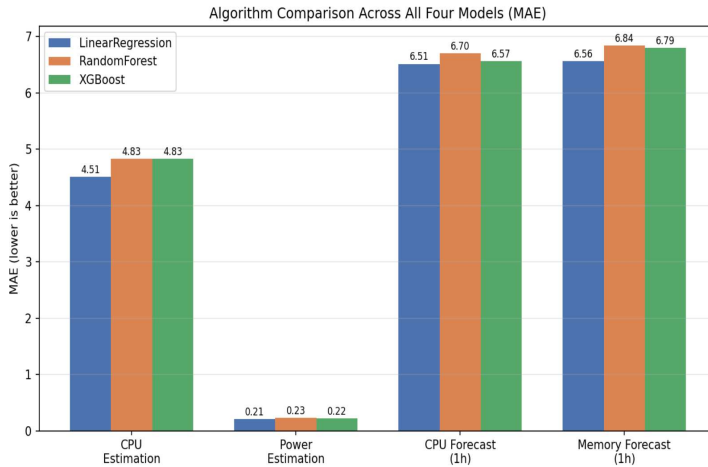
Both estimation models were trained on the real, 10,000-row “Green AI Data Center Telemetry” dataset (15 columns, with missing values only in `datacenter_region`, imputed with the most frequent category inside the preprocessing pipeline). Tables 5.1 and 5.2 report every candidate algorithm's result. In both cases Linear Regression was selected since neither Random Forest nor XGBoost cleared the pre-registered 5% relative MAE improvement bar over it. Figure 5.1 compares the MAE values across the evaluated models and provides a visual comparison of the model performance.

**Table 5.1 CPU Estimation Model: Candidate Comparison**  
(Target: cpu\_utilization)

| Model                        | MAE   | RMSE  | R <sup>2</sup> |
|------------------------------|-------|-------|----------------|
| Linear Regression (selected) | 4.511 | 7.671 | 0.845          |
| Random Forest                | 4.828 | 8.077 | 0.828          |
| XGBoost                      | 4.833 | 7.999 | 0.831          |

**Table 5.2 Power Estimation Model: Candidate Comparison**  
(Target: power, rescaled to 1.2–4.5 kW)

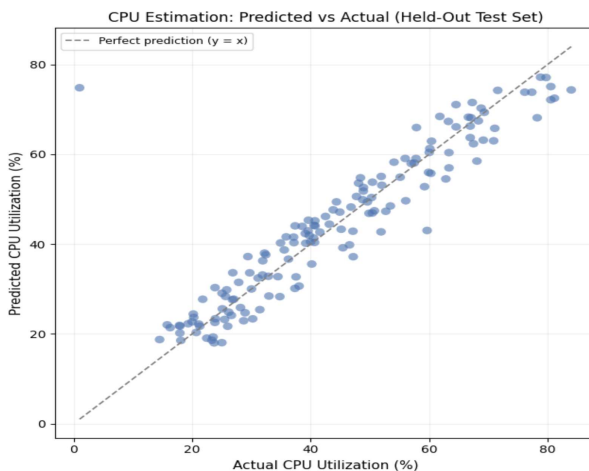
| Model                        | MAE   | RMSE  | R <sup>2</sup> |
|------------------------------|-------|-------|----------------|
| Linear Regression (selected) | 0.211 | 0.330 | 0.422          |
| Random Forest                | 0.227 | 0.346 | 0.366          |
| XGBoost                      | 0.222 | 0.341 | 0.385          |



**Fig 5.1. Comparison of evaluated algorithms across CPU estimation, power estimation, CPU forecasting and memory forecasting using MAE**



The CPU model explains the large majority of variance in the target ( $R^2 = 0.845$ ) using only workload and environmental features, with its own target correctly excluded to avoid leakage. The power model is noticeably weaker ( $R^2 = 0.422$  at best) even with `cpu_utilization` included as an input. This is a genuine and expected finding rather than a bug and indicates that power draw in the real dataset depends on hardware and efficiency factors not fully captured by the available feature set. This asymmetry between the two targets is consistent with the general pattern reported in the wider literature on data-center power modelling (Section 2.4) and directly informed the decision to make the power model's prediction the primary basis for the what-if energy estimate, with a simpler CPU-ratio formula used only as a fallback when the model or the target's cooling telemetry is unavailable. Every number produced this way is explicitly labelled as an estimate rather than presented as a measurement. Figure 5.2 further illustrates this performance through the predicted-versus-actual plot, where the predictions show a close correspondence with the actual CPU utilization values across the held-out test set.



**Fig. 5.2. CPU Estimation: Predicted vs Actual**

### **5.3 WHAT-IF SIMULATION FOR SERVER CONSOLIDATION**

Consider server S6 (Compute, workload bias 0.05, “very idle”), which is reliably flagged `idle_server` once its six-hour rolling average CPU falls below the Compute-type threshold of 17.3%. Requesting a what-if simulation for S6 causes the engine to enumerate every other Compute-type server not itself `idle-flagged`. In this fleet that means S1, S2 and S13. The engine computes each one's post-move CPU, memory and network as its own current load plus 90% of S6's current load. Each candidate is then checked against the four safety dimensions in Table 3.7. A candidate already running close to its own capacity or whose predicted post-move inlet temperature would exceed 30°C is excluded from `safe_now` regardless of headroom on other dimensions. Among the remaining candidates, the one with the most spare headroom after the move is chosen as the recommended target and its own fifteen-minute-ahead CPU and memory forecast is checked in exactly the same way before the recommendation is finalized as `safe`. The resulting recommendation card shows the chosen target, the full list of considered candidates with their pass/fail status on each dimension and an estimated facility-power impact derived from the trained power model, with the estimate's assumptions (which model was used and that it is a legitimate estimate rather than a measurement) stated explicitly alongside the number, consistent with the honesty principle followed throughout the system.

### **5.4 HONEST SUMMARY OF SYSTEM MATURITY**

Consistent with the principle of never presenting an estimate with the same confidence as a measurement, the current system can be summarized in three honest categories.

#### ***5.4.1 GENUINELY REAL***

The rule engine (pure threshold logic, no ML involved), PUE (a direct ratio of two independently simulated values), the offline ML validation results reported in Tables 5.1 and 5.2, the timestamp-based energy integration, and the four-dimension consolidation safety check.

#### ***5.4.2 LEGITIMATE ESTIMATES, CLEARLY LABELLED***

Cost, carbon, water, and the power model's what-if energy prediction, which agrees with the simulator on direction and scale but will not match it exactly, since the two are independently built approximations of the same underlying quantity.

#### ***5.4.3 DELIBERATELY NOT BUILT***

Criticality and SLA-aware gating, target storage-capacity and electrical/PDU-capacity safety checks, a verified (rather than operator-attested) confirmation mechanism for executed actions, the CPU model's designed hardware-health-check role (which needs environmental-fault injection in the simulator first), and trained six-hour/twenty-four-hour forecast models, for which there is not yet enough real telemetry history.

## 5.5 SYSTEM IMPLEMENTATION

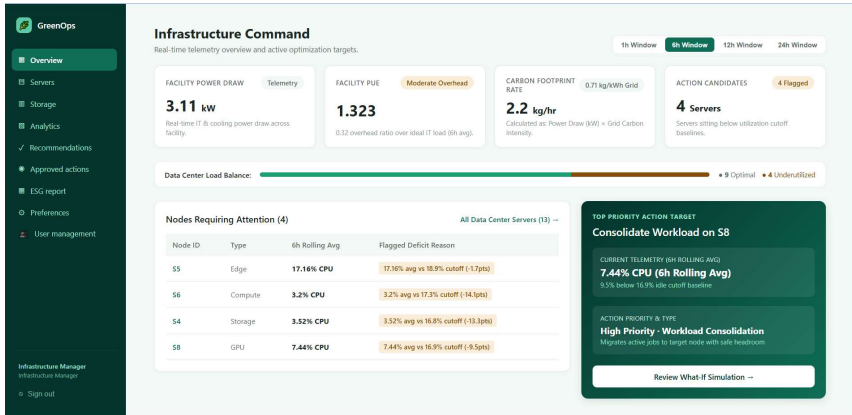


Fig. 5.3. Dashboard Overview Page

The Overview page summarizes facility wide values for PUE, cumulative energy, carbon and the current count of flagged servers being refreshed from live simulated telemetry.

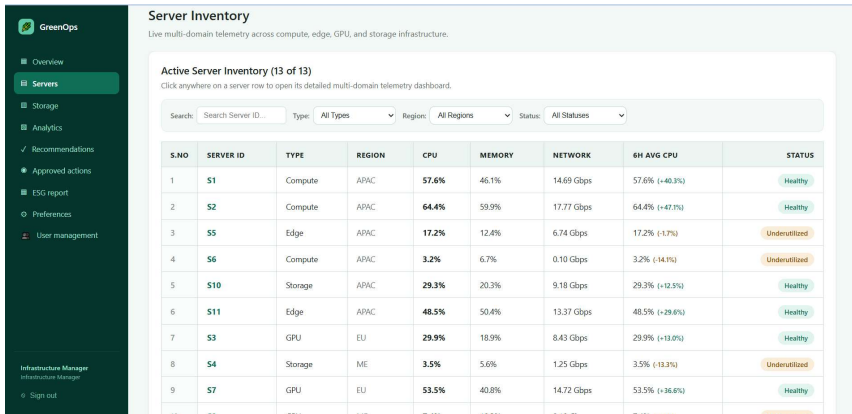


Fig. 5.4. Servers Details Page

The Server detail page gives insights into per-server view combining live environmental reading, the CPU forecast and any active waste flags for that specified server.

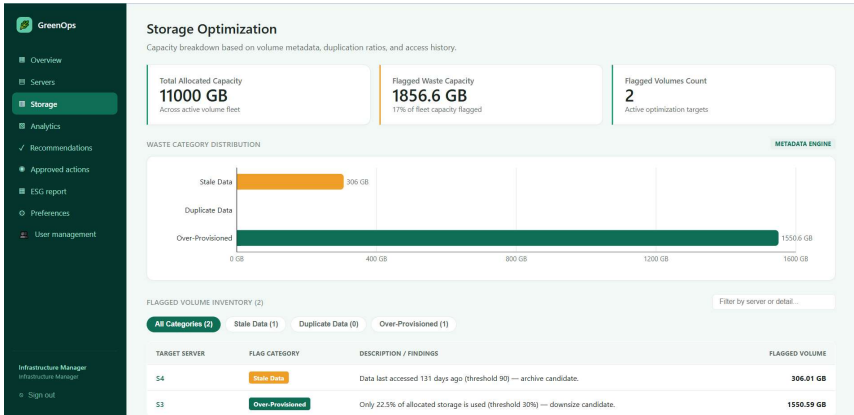


Fig. 5.5. Storage Details Page

The Storage page shows specific waste view, surfacing stale, duplicated and over-provisioned volumes flagged by the rule engine.

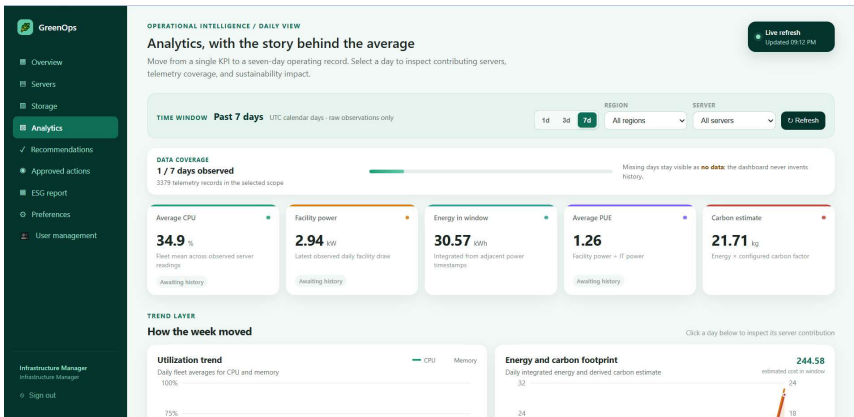


Fig. 5.6. Analytics Page

The Analytics page breaks down the facility-level of idle servers, power/cooling performance and storage waste over time which is used to spot trends.

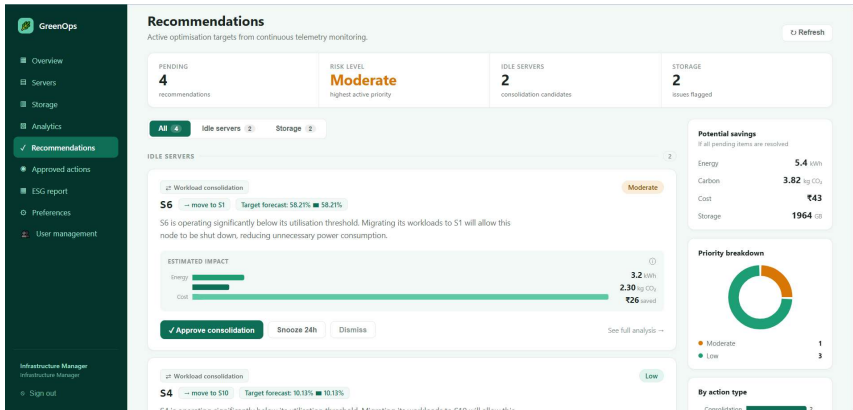


Fig. 5.7. Recommendations Page

The Recommendations page ranks a list of pending consolidate/rightsize actions with inline Accept / Snooze / Dismiss controls, mini impact charts and a sidebar summarizing pending savings across the batch.

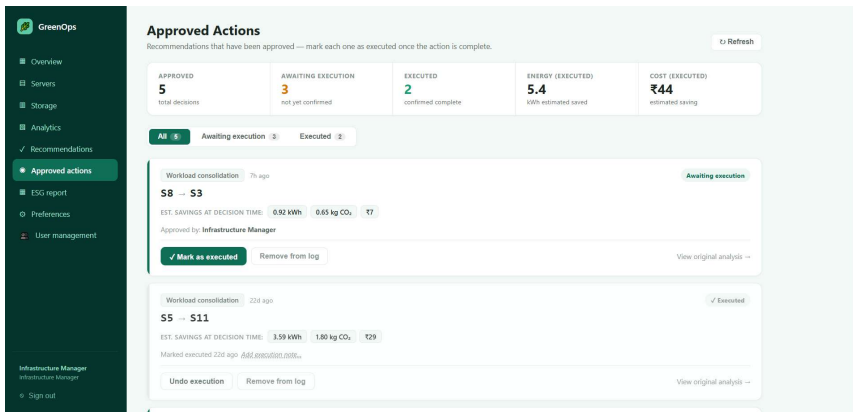
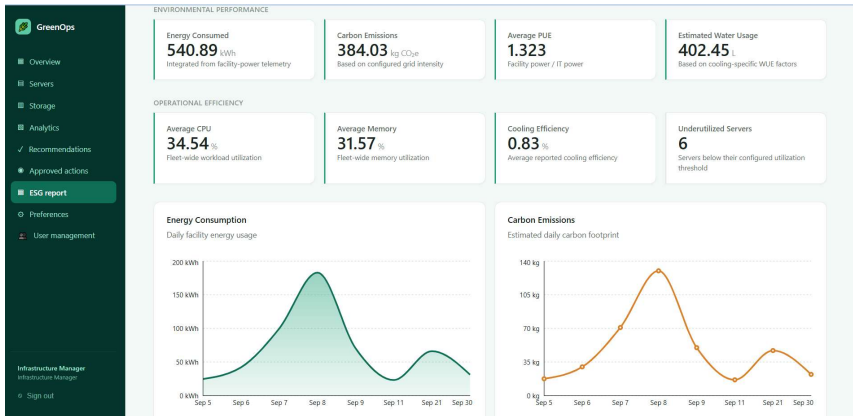


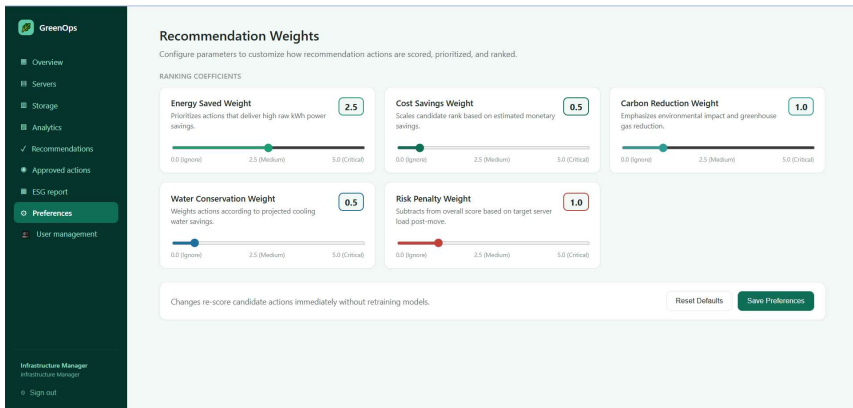
Fig. 5.8. Approved Actions Page

The Approved actions logs the past operators decisions, filterable by executed vs awaiting executions where an operator can retroactively attach an executed at timestamp and a note once the action is carried out.



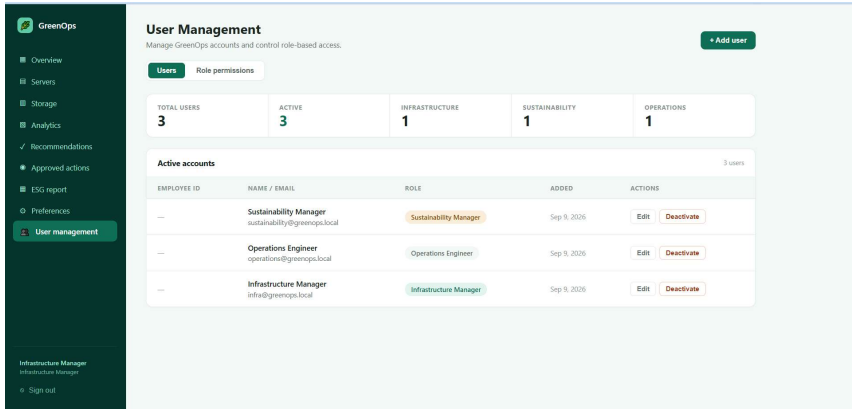
**Fig. 5.9. ESG Reporting Page**

This page has aggregated log of operator-action over a selected date range, exportable as CSV or PDF, presented explicitly as a decision record.



**Fig. 5.10. Preferences Page**

The Preferences page has a weight slider for cost, carbon, water and risk that feeds directly into the recommendation scoring formula, re-ranking pending recommendations instantly without retraining any model.



**Fig. 5.11. User Management Page**

The User Management page provides administrative controls for managing GreenOps accounts and role-based access.



## **CHAPTER 6**

### **CONCLUSION AND FUTURE WORK**

#### **6.1 CONCLUSION**

In this project, we have discussed the problem of inefficient resource utilization and sustainability in on-premise data centers and have developed GreenOps, a decision-support platform that would help in monitoring infrastructure resources and identifying possible optimization opportunities. The proposed system combines telemetry monitoring, sustainability calculation, workload forecasting and recommendation generation to provide a unified solution for managing underutilized servers in on-premise environments. It has been noted that the functioning of the system involves multiple stages, starting from telemetry collection and identification of underutilized resources, followed by workload prediction and what-if analysis before providing a recommendation to the operator. The system calculates important sustainability measures such as energy, carbon, water and cost, along with PUE and WUE, and uses statistically derived thresholds to identify inefficient resource usage. The implemented models achieved an  $R^2$  value of 0.845 for CPU estimation and 0.422 for power estimation. The recommendation engine further allows different priorities to be assigned to sustainability and performance factors before suggesting suitable actions. The proposed system does not automatically execute any infrastructure changes, and the final decision remains with the human operator. Thus, GreenOps provides a transparent and priority-aware approach for supporting resource optimization and sustainability management in on-premise data center environments.

## **6.2 LIMITATIONS**

The proposed system provides a working platform for resource monitoring and optimization in on-premise data centers, but its real-world implementation poses certain challenges. The telemetry used in the current system is simulated rather than collected from actual data-center hardware, and the cooling simulation does not currently represent independent equipment failures. The cost, carbon and water calculations are also based on configured values rather than live external data. The what-if analysis does not yet consider certain factors such as storage capacity, electrical capacity, workload criticality and SLA requirements. In addition, longer-term forecasting for twenty-four-hour periods has not been implemented due to the limited historical data available, and approved actions are currently confirmed by the operator rather than independently verified. Storage optimization and ESG reporting also remain as areas for further development.

## **6.3 FUTURE WORK**

The proposed system can be further enhanced by replacing the simulated telemetry with real-time data from on-premise data-center infrastructure through sources such as Prometheus, hypervisor APIs, smart PDU and BMS systems. The cooling component can also be extended to support controlled environmental fault conditions, while live electricity tariffs, regional carbon-intensity information and actual water measurements can be incorporated to improve the sustainability calculations. The what-if analysis can be further developed by considering storage capacity, electrical capacity, workload criticality and SLA requirements before generating recommendations. With the availability of sufficient historical telemetry, a dedicated 24-hour workload forecasting model can also be developed to support longer-term resource optimization decisions. An independent verification mechanism can be

introduced to confirm approved actions and measure the actual savings achieved. The existing storage optimization and ESG reporting capabilities can also be further enhanced with additional analytics and reporting features, allowing GreenOps to provide a more comprehensive sustainability and resource-management solution for on-premise data centers.

## APPENDIX 1 - KEY GENERATING FORMULAS

The following formulas, reproduced from the simulator implementation, generate every live telemetry reading in the system.

### A1.1 SERVER\_MONITOR.PY

$$workload = clamp(0.02, 0.98, gauss(bias, 0.08))$$

$$cpu = clamp(1, 99, workload \times 100 + uniform(-5, 5))$$

$$memory = clamp(5, 99, cpu \times 0.9 + uniform(-8, 8))$$

$$network = max(0.1, workload \times 25 + uniform(-2, 2))$$

### A1.2 POWER\_MONITOR.PY

$$variable\_kw = 3.3 \times (0.85 \times cpu\_fraction + 0.15 \times memory\_fraction)$$

$$it\_power\_kw = max(0.3, 1.2 + variable\_kw + uniform(-0.15, 0.15))$$

$$penalty\_fraction = (0.99 - cooling\_efficiency) \div (0.99 - 0.4)$$

$$overhead\_ratio = max(1.05, 1.15 + 0.40 \times penalty\_fraction + uniform(-0.03, 0.03))$$

$$facility\_power\_kw = it\_power\_kw \times overhead\_ratio$$

### A1.3 COOLING\_MONITOR.PY

$$inlet\_temperature\_c = 19.0 + temp\_rise\_by\_type[type] \times cpu\_fraction + uniform(-0.8, 0.8)$$

$$cooling\_efficiency = clamp(0.4, 0.99, 0.95 - efficiency\_drop\_by\_type[type] \times cpu\_fraction + uniform(-0.03, 0.03))$$

where  $temp\_rise\_by\_type = \{Air: 9.0, Hybrid: 5.0, Liquid: 2.5\}$  °C at 100% CPU, and  $efficiency\_drop\_by\_type = \{Air: 0.35, Hybrid: 0.20, Liquid: 0.10\}$ .

#### **A1.4 CONSOLIDATION POST-MOVE LOAD**

$$post\_move\_cpu = target.current\_cpu + source.current\_cpu \times 0.9$$

$$post\_move\_memory = target.current\_memory + source.current\_memory \times 0.9$$

$$post\_move\_network = target.current\_network + source.current\_network \times 0.9$$

$$post\_move\_temp\_c = 19.0 + temp\_rise\_by\_type[target.type] \times \min(1.0, post\_move\_cpu / 100)$$

#### **A1.5 RECOMMENDATION RANKING SCORE**

$$score = \Sigma (weight_i \times normalized\_impact_i) - risk\_weight \times risk$$

## APPENDIX 2 - DATABASE SCHEMA

The following eleven tables make up the shared PostgreSQL database described in Section 3. Nullable columns are marked accordingly. Timestamp columns default to the moment of insertion unless otherwise noted.

### A2.1 USERS

| Column        | Type     | Notes                                                                 |
|---------------|----------|-----------------------------------------------------------------------|
| id            | Integer  | Primary key                                                           |
| employee_id   | String   | Unique, nullable                                                      |
| name          | String   | Not null                                                              |
| email         | String   | Unique, not null                                                      |
| password_hash | String   | infrastructure_manager, sustainability_manager or operations_engineer |
| is_active     | Boolean  | Default true                                                          |
| created_at    | DateTime | Default current time                                                  |

### A2.2 SERVERS

| Column            | Type    | Notes                       |
|-------------------|---------|-----------------------------|
| id                | Integer | Primary key                 |
| server_id         | String  | Unique, e.g. S1 through S13 |
| server_type       | String  | Default Compute             |
| cooling_type      | String  | Default Air                 |
| datacenter_region | String  | Default APAC                |

### A2.3 SERVER\_TELEMETRY

| Column                  | Type     | Notes       |
|-------------------------|----------|-------------|
| id                      | Integer  | Primary key |
| server_id               | String   | Indexed     |
| cpu_utilization         | Float    |             |
| memory_utilization      | Float    |             |
| network_throughput_gbps | Float    |             |
| workload_intensity      | Float    |             |
| timestamp               | DateTime | Indexed     |

### A2.4 POWER\_TELEMETRY

| Column            | Type     | Notes       |
|-------------------|----------|-------------|
| id                | Integer  | Primary key |
| server_id         | String   | Indexed     |
| it_power_kw       | Float    |             |
| facility_power_kw | Float    |             |
| timestamp         | DateTime | Indexed     |

### A2.5 COOLING\_TELEMETRY

| Column              | Type     | Notes       |
|---------------------|----------|-------------|
| id                  | Integer  | Primary key |
| server_id           | String   | Indexed     |
| inlet_temperature_c | Float    |             |
| cooling_efficiency  | Float    |             |
| cooling_type        | String   |             |
| timestamp           | DateTime | Indexed     |

## A2.6 STORAGE\_TELEMETRY

| Column                 | Type     | Notes       |
|------------------------|----------|-------------|
| id                     | Integer  | Primary key |
| server_id              | String   | Indexed     |
| total_storage_gb       | Float    |             |
| used_storage_gb        | Float    |             |
| duplicate_data_gb      | Float    |             |
| last_accessed_days_ago | Integer  |             |
| storage_type           | String   |             |
| timestamp              | DateTime | Indexed     |

## A2.7 FORECASTS

| Column              | Type     | Notes                                       |
|---------------------|----------|---------------------------------------------|
| id                  | Integer  | Primary key                                 |
| server_id           | String   | Indexed                                     |
| horizon_minutes     | Integer  | 60,360 or 1440 for the dashboard tier       |
| measured_cpu        | Float    |                                             |
| predicted_cpu       | Float    | Nullable when there is not enough history   |
| method              | String   | e.g. LinearRegression or historical_average |
| model_version       | String   | Nullable                                    |
| validation_mae      | Float    | Nullable                                    |
| validation_rmse     | Float    | Nullable                                    |
| validation_r2       | Float    | Nullable                                    |
| source_observations | Integer  | Default 0                                   |
| generated_at        | DateTime | Indexed                                     |



## A2.8 FLAGS

| Column       | Type     | Notes                                                      |
|--------------|----------|------------------------------------------------------------|
| id           | Integer  | Primary key                                                |
| server_id    | String   | Indexed                                                    |
| flag_type    | String   | idle_server, stale_data, duplicate_data or overprovisioned |
| severity     | String   | low, medium or high                                        |
| detail       | String   | Human-readable explanation                                 |
| metric_value | Float    | e.g. average CPU or GB reclaimable                         |
| created_at   | DateTime | Indexed                                                    |
| resolved     | Integer  | 0 active, 1 resolved                                       |

## A2.9 RECOMMENDATIONS

| Column              | Type     | Notes                                          |
|---------------------|----------|------------------------------------------------|
| id                  | Integer  | Primary key                                    |
| flag_id             | Integer  | Indexed, unique, not null                      |
| server_id           | String   | Indexed, not null                              |
| recommendation_type | String   | consolidate, archive, deduplicate or rightsize |
| priority            | String   | high, medium or low                            |
| explanation         | String   | Not null                                       |
| status              | String   | pending, accepted, rejected or snoozed         |
| created_at          | DateTime | Indexed                                        |
| decided_at          | DateTime | Nullable                                       |
| snoozed_until       | DateTime | Nullable                                       |

## A2.10 OPERATOR\_ACTIONS

| Column            | Type    | Notes             |
|-------------------|---------|-------------------|
| id                | Integer | Primary key       |
| recommendation_id | Integer | Indexed, not null |
| server_id         | String  | Indexed, not null |

| Column                         | Type     | Notes                                              |
|--------------------------------|----------|----------------------------------------------------|
| action                         | String   | consolidate, rightsize, snooze or do_nothing       |
| notes                          | String   | Nullable                                           |
| estimated_energy_saving_kwh    | Float    | Nullable, snapshot at decision time                |
| estimated_carbon_reduction_kg  | Float    | Nullable, snapshot at decision time                |
| estimated_cost_saving          | Float    | Nullable, snapshot at decision time                |
| estimated_storage_reclaimed_gb | Float    | Nullable, snapshot at decision time                |
| target_server_id               | String   | Nullable, the chosen consolidation target if any   |
| executed_at                    | DateTime | Nullable, null means approved but not yet executed |
| execution_note                 | String   | Nullable                                           |
| decided_by_user_id             | Integer  | Foreign key to users.id, nullable                  |
| decided_by_name                | String   | Nullable, snapshot at decision time                |
| decided_by_role                | String   | Nullable, snapshot at decision time                |
| created_at                     | DateTime | Indexed                                            |

## A2.11 PREFERENCES

| Column        | Type     | Notes                                 |
|---------------|----------|---------------------------------------|
| id            | Integer  | Primary key, single row by convention |
| energy_weight | Float    | Default 1.0                           |
| cost_weight   | Float    | Default 1.0                           |
| carbon_weight | Float    | Default 1.0                           |
| water_weight  | Float    | Default 0.5                           |
| risk_weight   | Float    | Default 1.0                           |
| updated_at    | DateTime | Updated on every write                |

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